Virutal Campus using 3D Scence Reconstruction

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STUDENTS: Haofei Zhao, Xinyu Yuan, Yingge Fan, Han Wu, Zhengwei Wang, Yuxuan Cong, Qiaowei Miao

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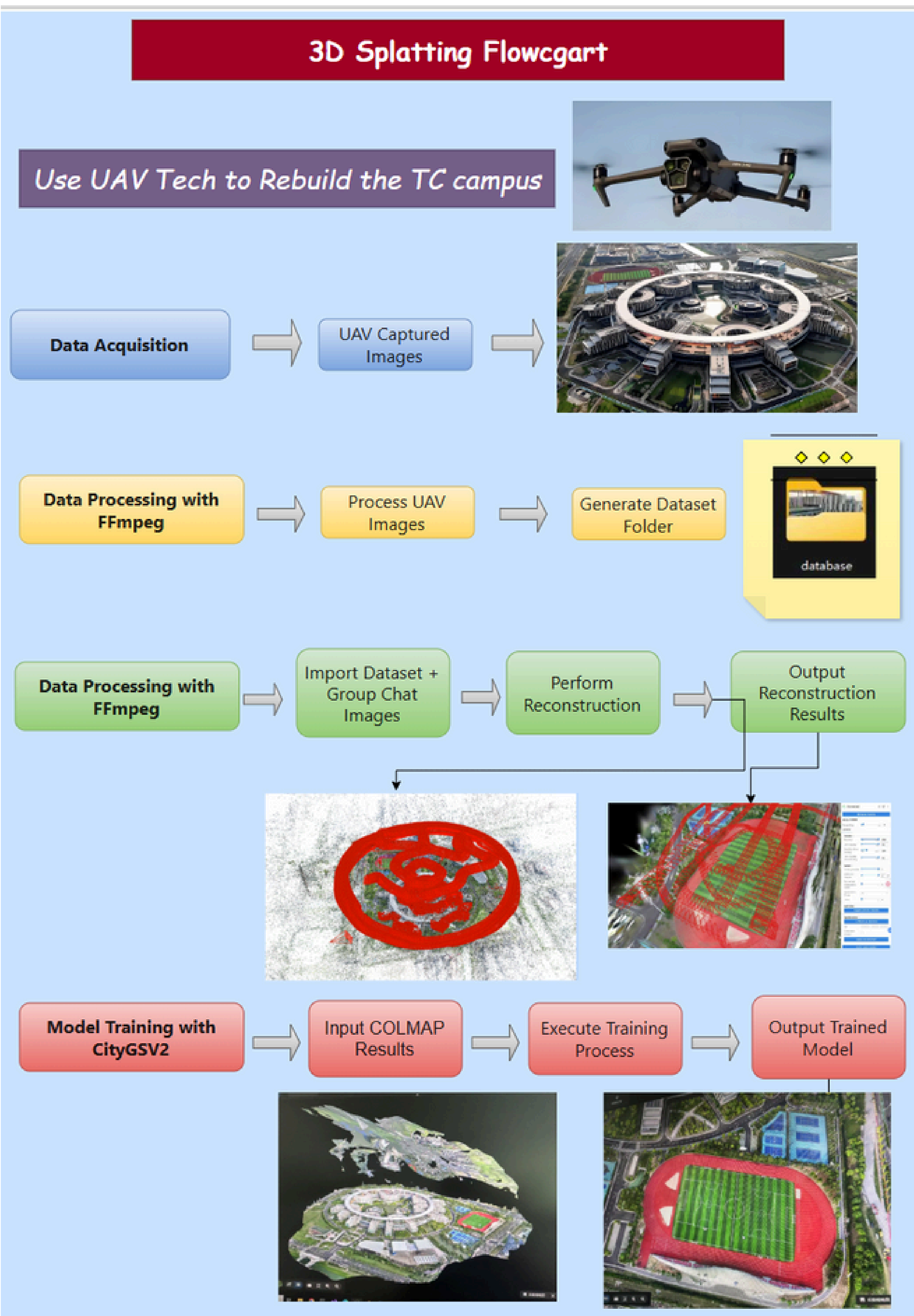
ABSTRACT

3D scene reconstruction is a rapidly advancing and impactful research area with applications in virtual reality, digital twins, and smart city planning. This project presents a Virtual Campus system based on photorealistic 3D reconstruction from aerial imagery. Utilizing drone-captured images and video, we reconstruct a high-fidelity virtual representation of a university campus using 3D Gaussian Splatting (3DGS)—a state-of-the-art technique that bypasses traditional meshing and texture baking.

INTRODUCTION

Digital twins of university campuses support a wide range of applications including navigation, outreach, and immersive teaching. However, traditional Structure-from-Motion (SfM) and Multi-View Stereo (MVS) pipelines—followed by mesh reconstruction—are often slow, fragile, and difficult to scale to large outdoor environments. In this project, we present a robust and efficient workflow for constructing a photorealistic virtual campus using 3D Gaussian Splatting (3DGS). Our pipeline employs UAV-based image and video capture, camera trajectory recovery with COLMAP, and scene reconstruction using CityGS/3DGS. Unlike mesh-based methods, 3DGS models the scene with view-dependent Gaussians, enabling real-time, high-fidelity rendering with rich detail and minimal artifacts. The resulting interactive virtual campus runs smoothly on consumer hardware and offers a scalable, reproducible solution for building campus digital twins.

METHDOLOGY



Data Acquisition and Preprocessing

We collected multi-view aerial videos of the XJTLU (Taicang) campus using a DJI Mavic 3 drone, following carefully planned flight paths to ensure comprehensive coverage. The captured videos were processed with FFmpeg to extract frames, which were then converted into images to create the dataset for 3D reconstruction. In total, we use 4000+ images to reconstruct the virtual campus.

Feature Extraction and Matching (COLMAP)

Feature Extraction: We employed the SIFT algorithm to detect keypoints in the images and compute corresponding feature descriptors.

Feature Matching: To ensure robust correspondence between image pairs, we applied the RANSAC algorithm to estimate the fundamental matrix, filter out mismatches, and retain geometrically consistent point pairs.

Sparse and Dense Reconstruction: Sparse reconstruction was used to generate a sparse 3D point cloud along with camera intrinsics and poses. Dense reconstruction further refined the results by producing a high-resolution dense point cloud.

3D Gaussian Splatting Modeling (3DGS/CITYGS V2)

We utilized CityGaussian V2 (CITYGS V2) as our core training model—a refined evolution of 3D Gaussian Splatting optimized for large-scale scene reconstruction. CITYGS V2 addresses convergence and scalability issues through techniques such as decomposed-gradient densification, depth regression, and an elongation filter, delivering improved geometric accuracy alongside substantial reductions in training time and memory usage—up to 25% faster training and 50% lower memory.

We fed our drone-captured campus imagery and COLMAP-derived outputs into CITYGS V2, then iteratively tuned Gaussian parameters to refine distribution quality and reduce reconstruction error. Training was accelerated using GPU-enabled novel-view synthesis, producing a final .ply point cloud embedding each Gaussian's 3D coordinates, RGB color, and density parameters. This output supports high-fidelity, interactive campus visualization on consumer hardware.

Experiment Setup and Challenges

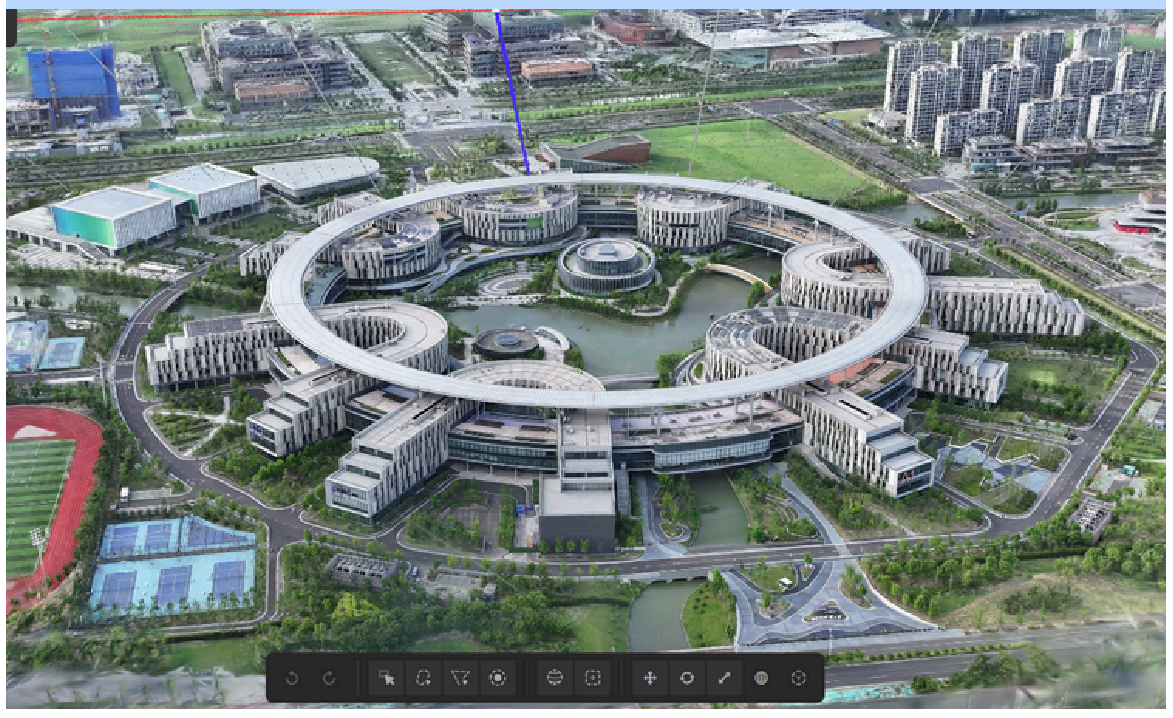
- **Setup:** CUDA/PyTorch/COLMAP version conflicts
  - pinned Conda env.
- **Poses:** Blur/sky/vegetation caused drift
  - frame filtering, more overlap, ground shots.
- **Training:** VRAM/time limits
  - regional training, fewer Gaussians/SH.
- **Alignment/Deploy:** GPS drift, heavy assets
  - marker alignment, LOD & compression.

RESULT

We successfully reconstructed multiple areas of our university campus using drone-captured video data. The raw video sequences were pre-processed with FFmpeg, which converted them into frame-by-frame images suitable for input into COLMAP. This process enabled accurate camera trajectory estimation and precise frame alignment.

By leveraging the 3D Gaussian Splatting (3DGS) framework, including the advanced CityGaussian pipeline, we generated highly detailed and photorealistic 3D reconstructions. The resulting virtual campus model supports smooth navigation, preserving fine details such as buildings, roads, and vegetation.

Compared to traditional mesh-based reconstruction methods and commercial software (e.g., MipMap), our approach demonstrated superior scene detail, higher visual fidelity, and enhanced real-time rendering capabilities.



Conclusion and Futurework

The Virtual Campus project successfully showcases the potential of 3D Gaussian Splatting (3DGS) for creating a highly realistic digital twin of the university campus, enriched by drone-based aerial imagery for comprehensive data capture. Evaluations from participants emphasize the platform's ability to seamlessly bridge physical and virtual spaces, enhancing accessibility for remote learners and enabling immersive interactions that simulate real campus life.

We gained valuable hands-on experience in 3D modeling, programming, and collaborative problem-solving, equipping them with essential skills for careers in technology and education.

Feedback from users highlights the educational value of the project, with praise for the intuitive interfaces and high-quality visualizations, which significantly enhance engagement and understanding.

Looking ahead, we plan to integrate augmented reality features to further enrich hybrid learning experiences and expand the virtual campus to include real-time collaborative tools, enabling global academic interactions.